
Modifying the Gravitational Constant and Hubble Parameter in a N-Body Simulation

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1 Introduction

Up until 1900 the notion of a static universe was widely accepted in the scientific community, with physicists such as Albert Einstein publishing work based on the static model [7] .

It was not until the early 1920s, when Alexander Friedmann and Georges Lemaître independently derived the Friedmann-Lemaître-Robertson-Walker equations [13] independently, which described an expanding universe.

With Hubble giving hard evidence to an expansion in 1929, this new model had been adopted by most scientists [1].

Within the scope of an expanding universe there are various theories, among which the big bang theory describes our early universe as extremely dense and hot, reaching temperatures of 10^{15} Kelvin 10^{-10} seconds into the universe's lifetime [10]. This may sound as if our cosmos is the result of an explosion, which is not the case. Instead the galaxies could be thought of as dots drawn on a balloon. As the balloon inflates, the distance between any two dots increases uniformly. In other words, to an observer in space all surrounding galaxies move radially away, no matter where the observer is positioned inside the universe. The discovery of the cosmic microwave background in 1965 validated the inflationary big bang theory. The cosmic microwave background (CMB) is electromagnetic radiation resulting from the big bang. It is almost isotropic, meaning that the intensity of radiation is nearly the same in any direction. It has a temperature of 2.7 [10]. The energy of photons relates to the wavenumber through $E = h/\lambda$, where h is the Planck constant and λ the wavelength. Thus, as space expands, so does the wavelength and the photon's energy decreases. The CMB was therefore much hotter in the early universe. Furthermore, by studying the anisotropy of the CMB one can determine the reduced matter densities of baryon, dark-matter, and other species and understand the formation of our galaxies [10].

If it were possible to solve a system of N bodies with arbitrary masses, initial position and velocities, interacting through the gravitational force, then one would have full knowledge of the phase space. In case of our universe this would enable us to accurately calculate the positions of galaxies at early times. Unfortunately, a system of more than three bodies can not be solved analytically (only in special cases when symmetries reduce the complexity)[11].

Rather than finding an analytical solution, the differential equations that govern the particle's positions and velocities can be approximated. Because these approximations deviate from the true solution, the time steps need to be small enough to a point, where these errors become negligible. For systems with many particles, the number of calculations would be close to impossible to solve by

hand. Due to recent technological advancements, however, this numerical limitation has been overcome with the use of computers.

By experimenting with different cosmological parameters (i.e. matter and radiation densities) physicists could now test for cosmological models that are similar to our universe [8].

Many of the simulation codes being used only support the Newtonian gauge as opposed to Einstein's proposed theory of general relativity (GR) [5]. It is a good approximation in many circumstances, although there are times when GR can not be ignored. At extremely large scales, the gravitational force vanishes. At galactic scales, highly relativistic black holes gain significance and at an early stage of our universe, a large amount of energy was radiation, which is highly relativistic as well.

Nonetheless, there are ways of incorporating relativistic corrections for a Newtonian simulation. In this thesis, we will explore the consequences of modifying variables in a Newtonian simulation by a time dependent constant, in search of new ways to incorporate a GR framework. By comparing the matter power spectra of simulations with various perturbations, we will outline interesting signatures that affect the formation of clusters and filaments and infer a model to describe the correlation between the change in power spectrum and perturbation.

2 Theoretical Introduction

There are various methods to account for relativistic effects. For instance, GR gauges may be defined such that Newtonian particles follow a relativistic trajectory. Making use of the linearized Einstein-Boltzmann equation in a dark-matter dominated era, one can infer equations of motion used in Newtonian simulations. While the simulation itself remains untouched, the outputted data is corrected for relativistic effects. For example by rotating the coordinate system to account for radiation. This is effective for large scale analysis [5]. It is inconsistent for smaller scales, where highly nonlinear gravitational collapse is increasingly important. Solving the coupled Einstein-Boltzmann equations without the linear approximation is currently not a viable approach for cosmological time frames.

The compromise between using a simulation code that supports general relativity, but is less efficient and using a Newtonian simulation, but merely manipulating the resulting data may be to implement minor changes to the Newtonian simulation. Two parameter that can be changed are the Hubble parameter and the gravitational constant.

2.1 Solutions of a Homogeneous and Isotropic System

In an expanding, homogeneous and isotropic system, particles move radially away from each other at the speed

$$\mathbf{v} = a(t)\mathbf{r} \quad (2.1)$$

where a is the scale factor that changes in time. Therefore it is useful to work with comoving coordinates defined in equation 2.2

$$r(t) = a(t)x \quad (2.2)$$

where x is constant in time .

The scale factor is determined by the following relation.

$$\mathcal{H}^2 + \frac{k}{a^2} = \frac{8\pi}{3}G\rho + \frac{\Lambda}{3} \quad (2.3)$$

Which was derived by Friedmann in the 1920s [1]. where p is the pressure, G is the gravitational constant, ρ is the energy density, and $\mathcal{H} = \dot{a}/a$ is the Hubble parameter. The curvature is given by k and has the values ± 1 or 0 which determines the cosmological paradigm. A positive curvature results in an ever expanding universe, while negative curvature leads to the universe collapsing in on itself. The threshold lays at $k = 0$.

Λ refers to the cosmic constant and is the vacuum energy density. $\Lambda > 0$ causes

a repulsive force that counteracts gravity. A model with a positive cosmic constant could retain an open universe even with negative curvature. Λ has been theorized by Einstein who found it necessary for a static universe to exist [10].

ρ is time dependent and made up of various particle species (photons, baryons, neutrinos, dark matter, dark energy). These species evolve differently in time. In a flat universe where the radiation density is negligible and $\Lambda > 0$, the scale factor according to the Friedmann equation 2.3 can be is in 2.4. This is a good approximation of our universe in a .

$$a^3 = \frac{\rho_{m,0}}{2\rho_{\Lambda,0}} \left[\cosh \left(\sqrt{3\Lambda} t \right) - 1 \right] \quad (2.4)$$

Where

- $\rho_{m,0}$ is the matter density at present time
- $\rho_{\Lambda,0} = \frac{\Lambda}{8\pi G}$

In the case of an empty universe ($\rho_m = 0$) the scale factor is proportional to 2.5

$$a \propto \exp \left(\left[\frac{1}{3}\Lambda \right]^{\frac{1}{2}} t \right) \quad (2.5)$$

The model used during this thesis is the Λ -cdm model, which postulates only three components making up the energy density: Dark energy, cold dark matter and baryonic matter. This means that contributions through other factors are not considered. The Friedmann equation can then be rewritten as 2.6

$$\mathcal{H} = \mathcal{H}_0 \sqrt{(\Omega_{cdm,0} + \Omega_{m,0})a^{-3} + \Omega_{\Lambda,0}} \quad (2.6)$$

Where

- $\mathcal{H}_0$, the Hubble parameter at present time
- $\Omega_{m,0}$ is the baryonic matter contribution to the total energy density at the present time and is set to 0.048
- $\Omega_{cdm,0}$ is the cold dark matter contribution and is set to 0.264
- $\Omega_{\Lambda,0}$ is the dark energy contribution and is set to 0.689

[12], [1]

2.2 Primordial Density Fluctuation

In our universe, the homogeneity and isotropy are disturbed by small fluctuations in the matter density.

Soon after the big bang, several processes caused large scale perturbations to

form quantum fluctuations. The primary anisotropy is due to the acoustic oscillations in the primordial plasma (about 380,000 - 487,000 years after the big bang). Initially gravity and outward pressure were competing forces in the formation of larger scale structures, through which acoustic oscillations would propagate through space carrying matter. Once the universe cooled off enough for atoms to form, photons were no longer trapped by electrons and protons, releasing pressure and enabling larger structures to form through gravity. [10], [14] .

2.3 Newtonian Cosmology

With the equations for an expanding, homogeneous, and isotropic system at hand one can reevaluate the equations of motion in a comoving coordinate system, which will be used in the simulation.

In N-body simulations, the particles trajectories are updated according to the three equations 2.7 - 2.9, which are the Poisson equation and equations of motion respectively [9].

$$\nabla^2\Phi = 4\pi G\rho - \Lambda \quad (2.7)$$

$$\frac{d\mathbf{r}}{dt} = \mathbf{u} \quad (2.8)$$

$$\frac{d\mathbf{u}}{dt} = -\nabla\Phi \quad (2.9)$$

With

- G is the gravitational constant
- ρ is the matter density
- Φ is the potential
- Λ is the cosmic constant

G the gravitational constant, ρ the matter density and Φ the potential. Notice that the Poisson equation takes the same form as the coulomb potential if Λ were zero, since they are both central potentials.

Next we will introduce new variables in comoving space in equations 2.10 - 2.12.

$$\mathbf{v} = \mathbf{u} - \mathcal{H}\mathbf{x} \quad (2.10)$$

$$\tilde{\Phi} = \Phi + \frac{\ddot{a}r^2}{a} \quad (2.11)$$

$$d\tau = \frac{1}{a}dt \quad (2.12)$$

Where τ is the conformal time and $\mathbf{v}$ is the velocity of particles in comoving coordinates.

In the new coordinate system the equations of motion can then be rewritten as

$$\frac{d\mathbf{x}}{d\tau} = \frac{1}{a}(a\mathbf{v}) \quad (2.13)$$

$$[\partial_\tau + \mathcal{H}]\mathbf{v} = -a\nabla\tilde{\Phi} \quad (2.14)$$

Using the relationship in 2.6, the Poisson equation in comoving coordinates is

$$\nabla^2\tilde{\Phi} = 4\pi G\rho_0 a^{-1}\delta \quad (2.15)$$

Where $\delta = (\rho - \bar{\rho}) / \bar{\rho}$ is the overdensity and ρ_0 is the matter density at $t = 0$ [9], [4].

One way of accounting for relativistic affects on smaller scales may come from manipulating the parameters that occur in equations 2.15 - 2.14. More precisely a time dependent perturbation can be added to the Hubble parameter in the Euler equation 2.16 and the gravitational constant in the Poisson equation 2.17, as in

$$\mathcal{H} \longrightarrow \mathcal{H} + a\delta\mathcal{H} \quad (2.16)$$

$$G \longrightarrow G + a\delta G \quad (2.17)$$

Furthermore, we introduced this change at different times during the simulation. The effect that equation 2.16 has, becomes more apparent when looking at the definition of the Hubble parameter:

$$\dot{a} = a(\mathcal{H} + a\delta\mathcal{H}) \quad (2.18)$$

At later times, when Ω_Λ dominates in 2.6 $a(t)$ is proportional to

$$a(t) \propto \left[\left(1 + \frac{\delta\mathcal{H}}{\sqrt{\Lambda/3}} \right) \exp\left(-\sqrt{\frac{\Lambda}{3}}t\right) - \frac{\delta\mathcal{H}}{\sqrt{\Lambda/3}} \right]^{-1} \quad (2.19)$$

Therefore the universe expands faster than for an unperturbed system, as can be seen in equation 2.5.

2.4 The Power Spectrum

The power spectrum is a common tool to analyze cosmological quantities such as metric potentials or matter density [8]. In this thesis we will be focusing on the matter power spectrum $P(k)$, which is defined as

$$\xi(r) = \frac{1}{V} \int d^3\mathbf{x} \delta(\mathbf{x}) \delta(\mathbf{r} - \mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} P(k) e^{i\mathbf{k}(\mathbf{x}-\mathbf{x})} \quad (2.20)$$

with $\xi(\mathbf{r})$ being the matter correlation function $k = |\mathbf{k}|$ [1]. The power spectrum quantifies the contribution of each wavenumber k to the overdensity. Inferred from cosmic experiments (using the cosmic microwave background anisotropies, weak lensing of galaxy shapes, and more) the power spectrum of our current universe could be extrapolated and can be seen in Figure 2.1. For small wave-

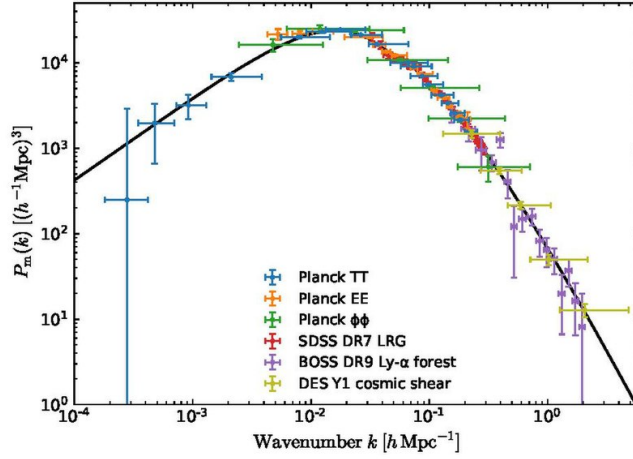


Figure 2.1: Power spectrum of our current universe [3]. The theoretical curve (black line) matches the experimental data quite well.

lengths $P(k)$ is proportional to k , while for large wavelengths it is proportional to k^{-3} and has a maximum at $k \approx 2 \times 10^{-2} [\frac{h}{Mpc}]$. The range that will be looked at in the thesis is from $10^{-1} [\frac{h}{Mpc}]$ to $10^1 [\frac{h}{Mpc}]$, where nonlinear, gravitational collapse becomes increasingly important.

3 Simulation Code

Computers can complete up to 3 billion operations per second. Comparing this to a human's ability to do calculations, one can easily see the potential in solving physical systems computationally. N-body systems have been used, since the advent of computers in areas such as plasma simulations, compressible fluid dynamics and cosmology.[9]

A common N-body simulation is *gevolution* [4] and will be used for the duration of this thesis. The simulation code of *gevolution* begins with initializing the particles. As discussed earlier, due to primordial density fluctuations amplified through gravitational attraction, our universe is inhomogeneous and anisotropic. This translates into small shifts in position and velocity of particles in the simulation. The Cosmic Linear Anisotropy Solving System (CLASS) code calculates the density and velocity transfer function, which determines the average magnitude and direction of these shifts [6]. A random number generator returns each particle's perturbation within the constraints of the transfer function. The seed parameter in *gevolution* ensures that any two simulations with the same seed yield identical results. This makes comparing two simulations with different perturbations more accurate, since statistical fluctuations are correlated. It is crucial to match the cosmic parameters (especially the Newtonian gauge) for CLASS and the N-body simulation to ensure the accuracy of the transfer functions. The initialization of particles is followed by the main loop of the simulation.

The simulation code cycles through three steps, each step incorporating one of the equations 2.14 - 2.15.

3.1 Poisson Equation

Firstly ρ is calculated. Due to its efficiency, most cosmological codes take advantage of a simple cubic lattice structure. A particles mass is partitioned onto the neighboring eight lattice points, as done by the cloud-in-cell method [9]. The mass assigned to lattice points depends on the displacement vector to the particle according to

$$\Delta m = C \cdot (1 - d_x d_y d_z) \quad (3.1)$$

Where $0 < d_i < 1$ is the distance between the particle and the neighboring particles in Cartesian coordinates and C is a normalizing constant, ensuring that the partitioned mass remains the same.

The cloud-in-cell method is iterated over all particles in the simulation. Particles at the border of the lattice are assigned to the nodes on the opposite side of the lattice to ensure periodic boundaries.

Because the particle masses are distributed along a lattice grid, the Poisson equation becomes computationally economical to solve in the reciprocal lattice. A scalar field on the reciprocal lattice can be derived according to

$$\hat{f}(\mathbf{G}_m) = C \sum_n f(\mathbf{R}_n) e^{-i\mathbf{R}_n \mathbf{G}_m} \quad (3.2)$$

Where f is the scalar field on the lattice and the relative coordinates of the lattice points $\mathbf{R}_n$ are defined as

$$\mathbf{R}_n = n_1 \mathbf{e}_1 + n_2 \mathbf{e}_2 + n_3 \mathbf{e}_3$$

and the coordinates on the reciprocal lattice are given by

$$\mathbf{G}_m = m_1 \mathbf{e}_1 + m_2 \mathbf{e}_2 + m_3 \mathbf{e}_3$$

where n_i are in units such that the distance between adjacent nodes is equal to 1. The Fourier transform of the Poisson equation is

$$\hat{\Phi}(\mathbf{G}_m) = G(\mathbf{G}_m) \hat{\rho}(\mathbf{G}_m) \quad (3.3)$$

G is the green function in Fourier space defined as

$$G = \frac{3\pi G}{a} \left[\sum_{i=1}^3 \sin\left(\frac{m_i}{2}\right)^2 \right]^{-1} \quad (3.4)$$

Finally $\hat{\Phi}$ is transformed back into real space. [9], [4]

3.2 Equations of Motion

The gradient of Φ at a particle's position is constructed similarly to the matter assignment by interpolating the gradient fields of neighboring nodes and multiplying by a weight that depends on the distance between the particle and nodes. The x component of the gradient is calculated as per equation 3.5.

$$\begin{aligned} \nabla\Phi(\mathbf{x})_x = & (1 - d_y)(1 - d_z) \cdot [\Phi(\mathbf{R}_n + \mathbf{e}_x) - \Phi(\mathbf{R}_n)] + \\ & d_y(1 - d_z) \cdot [\Phi(\mathbf{R}_n + \mathbf{e}_y + \mathbf{e}_x) - \Phi(\mathbf{R}_n + \mathbf{e}_y)] + \\ & (1 - d_y)d_z \cdot [\Phi(\mathbf{R}_n + \mathbf{e}_z + \mathbf{e}_x) - \Phi(\mathbf{R}_n + \mathbf{e}_z)] + \\ & d_y d_z \cdot [\Phi(\mathbf{R}_n + \mathbf{e}_y + \mathbf{e}_z + \mathbf{e}_x) - \Phi(\mathbf{R}_n + \mathbf{e}_y + \mathbf{e}_z)] \end{aligned} \quad (3.5)$$

Where d_i is the distance between particle and lattice point in units of the lattice constant and $\mathbf{e}_i$ is the displacement vector between two neighboring lattice points. The other components are calculated similarly.

The velocity and position are updated using the leapfrog integration, an explicit numerical integrator. It is used to numerically approximate the velocity and position, when the potential only depends on the positional vector. The leapfrog integration is a popular technique, as the phase space coordinates can be calculated for both directions in time by rearranging the equation. Therefore, the time invariance guarantees conservation of energy.

Moving on, equation 2.14 can be rewritten as the following

$$\frac{d(a\mathbf{v})}{d\tau} = -a\Phi \quad (3.6)$$

Through the leapfrog integration, $a\mathbf{v}$ is updated according to

$$a\mathbf{v}(\tau + 0.5 \Delta\tau) = a\mathbf{v}(\tau - 0.5 \Delta\tau) - a(\tau)\nabla\Phi(\mathbf{x}\tau)\Delta\tau. \quad (3.7)$$

After updating the velocity, the scale factor is evaluated at time $\tau+0.5\Delta\tau$ during the simulation. Since $da / d\tau = \mathcal{H}(a) a^2$ is a first-order, ordinary differential equation, the scale factor can be calculated with the 4th order Runge-Kutta-Method, which essentially evaluates $\dot{a}$ at different points between τ and $\tau+0.5\Delta\tau$ to evolve a 3.8.

$$\begin{aligned} k_1 &= \frac{\Delta\tau}{2} a^2 \cdot \mathcal{H} \\ k_2 &= \frac{\Delta\tau}{2} (a + k_1/2)^2 \cdot \mathcal{H}(a + k_1/2) \\ k_3 &= \frac{\Delta\tau}{2} (a + k_2/2)^2 \cdot \mathcal{H}(a + k_2/2) \\ k_4 &= \frac{\Delta\tau}{2} (a + k_3)^2 \cdot \mathcal{H}(a + k_3) \\ a(\tau + 0.5 \Delta\tau) &= a(\tau) + (k_1 + 2k_2 + 2k_3 + k_4)/6 \end{aligned} \quad (3.8)$$

This will become important when applying the perturbation to the Hubble parameter as we will see further on. RK4 has an error of $\mathcal{O}(\Delta\tau^4)$, so $\Delta\tau$ has to be chosen small enough [1].

Subsequently, the particle's position can be updated using the relation in 2.13 to acquire the updated position according to

$$\mathbf{x}(\tau + \Delta\tau) = \mathbf{x}(\tau) + \Delta\tau\mathbf{v}(\tau + 0.5\Delta\tau) \quad (3.9)$$

Finally, the scale factor evolves by half a conformal time step with 3.8 and the cycle is repeated. The power spectra are calculated at predefined redshifts during the simulation. [4]

3.3 Modifying the Simulation Code

The modification of the gravitational constant results from changing the Poisson equation. This is done by substituting 2.17 in 3.3. Although the potential is calculated on the reciprocal lattice, this does not affect the gravitational constant, since it does not depend on the positional vector. It is important to note that the gravitational constant as per implementation by *gevolution* is proportional to the simulation length squared, which needs to be accounted for when adding the time dependent factor for varying simulation sizes.

The modifications to the Hubble parameter is slightly more complicated. The scale factor is updated each cycle by solving the ordinary differential equation $da/d\tau = a^2\mathcal{H}$. Next, by substituting 2.16 for the Hubble parameter each derivative k_i in the RK4 method 3.8 is modified by an additional term, which for k_1 is the following:

$$k_1 \longrightarrow k_1 + \frac{\Delta\tau}{2}a^3 \cdot \delta\mathcal{H}$$

Since the Hubble parameter is proportional to the square root of gravitational constant, this needs to be considered when using various simulation lengths. Lastly we introduced a variable for both parameters to toggle the modifications depending on the redshift of the simulation.

3.4 Parameter Setup

This paper is intended as a guide for more in depth analysis to come, which is why it is not necessary to conduct large simulations, as done by Julian Adamek in [8]. Nevertheless, it is important to avoid systematic errors caused by insufficient particle numbers or lattice units.

Firstly, the power spectrum is calculated by averaging the amplitudes of the modes on a shell in the reciprocal lattice. Consequently, the power spectra is only accurate up to the Nyquist frequency $N_G/(2L)$, where L is the length of the lattice. For this reason we chose the number of lattice points to be 256^3 .

By appending the power spectra of a smaller scale simulation to one of a larger scale, we can circumvent the need for a simulation with more particles. To obtain the matter power spectrum for wavelengths between $10^{-1}[\frac{h}{Mpc}]$ and $10^1[\frac{h}{Mpc}]$ we implemented two different simulation lengths that can be seen in 3.4. Lastly,

the number of particles influenced the structure formation, meaning that the power spectra of an unperturbed simulation will undergo a systematical shift from the theoretical curve. This only occurred when the number of particles or lattice points was insufficient. To avoid this error and also avoid simulations that are too large, the number of particles is 256^3 for all simulations.

- Lattice points per dimension: 256
- Number of particles: 256^3
- Simulation size in $[\frac{Mpc}{h}]$: 150, 28

All of the remaining parameters used in the simulation are shown in Table A.1.

4 Comparing the Unperturbed Simulation

4.1 Comparing Different Seeds

Since the initial displacements of the particles are generated at random, the simulation is prone to statistical errors that diminish with increased particle numbers. To quantify this I looked at the relative difference of the PS with different seeds

$$(Pk_{1234} - Pk_{42}) / Pk_{42}$$

, where the indices refer to the seed that was used. The results are shown in Figure 4.1. The red data points are from the larger simulation with length $150 \left[\frac{M_{ps}}{h} \right]$ and the green data points are from the smaller simulation with length $28 \left[\frac{M_{ps}}{h} \right]$.

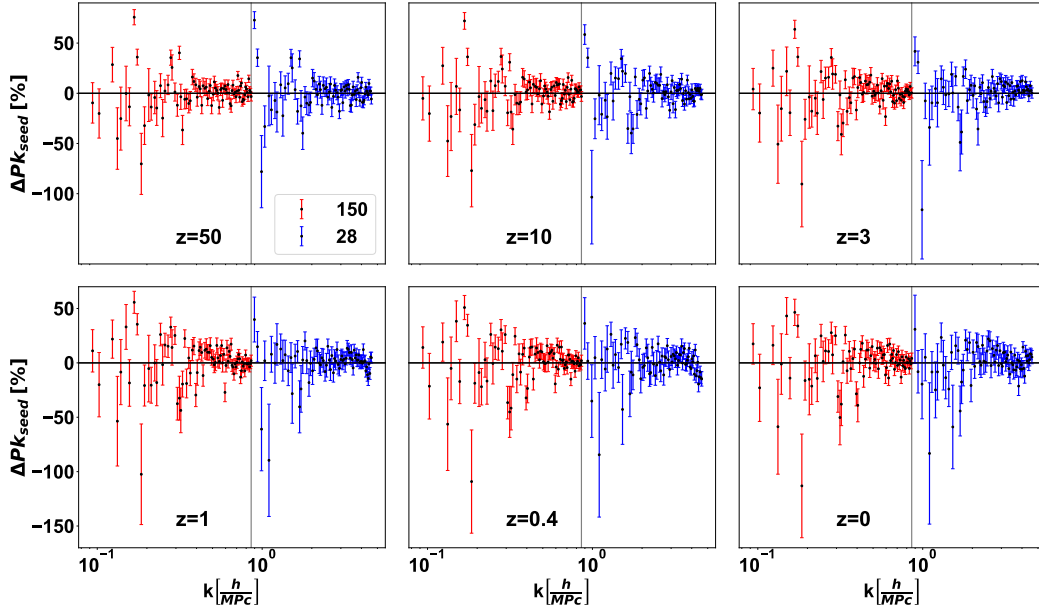


Figure 4.1: relative difference of matter power spectra for different seeds, shown at different redshifts. The colors refer to the simulation size. The data points vary up to 100 %

Most data points are within one standard deviation away from zero. The power spectra are furthest apart at small wavenumbers with relative differences of up to 100 % and approach each other as the wavenumber increases. This can be seen for both simulation sizes. A reason for the change in uncertainty is that, as discussed earlier, the power spectrum is derived from the number of particles inside a finite shell on the reciprocal lattice. As the shell decreases in radius so does the sample size, which results in a larger statistical uncertainty.

Apart from the statistical uncertainty a systematical trend can be seen for

smaller redshifts especially at larger k . The most likely cause is the particles displacement at the beginning of the simulation. Since the perturbation in position and velocity is generated randomly (within the limitations of the transfer function), the matter field could be more homogeneous for one simulation than for another at small scales. Since gravity tends to enhance inhomogeneity, the initially small difference could increase as the simulation progresses. If the number of particles approaches infinity, these initial differences would average out to zero. Therefore, this error is tied to the number of particles in the simulation. At first glance, relative differences of up to 100% may sound alarming, but if the same seed is used the statistical uncertainty is correlated. Thus, the ratio of two power spectra yields a far smaller statistical uncertainty.

4.2 Comparison to CLASS

To verify that the chosen settings yield an accurate result it can be beneficial to compare the power spectrum to the ones calculated by CLASS. In addition to the linear treatment CLASS can approximate the non linear corrections. The quotient of the matter power spectrum inferred from the *gevolution* simulation code and the power spectrum calculated with the non linear approximation by CLASS are shown in Figure 4.2.

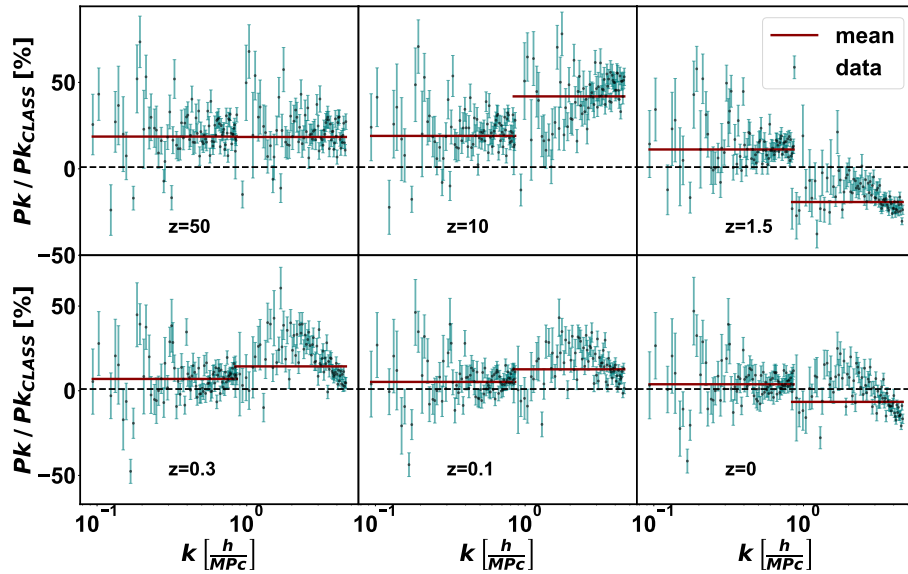


Figure 4.2: Relative difference between power spectrum generated by *gevolution* and the power spectrum calculated through CLASS

The power spectra are in different coordinate systems, thus the data points

from *gevolution* first needs to be multiplied by a^6 before it can be divided by the data points from CLASS. Additionally to the data point one can see the average value for each simulation. To account for the error on every data point the mean was calculated as a weighted average.

$$\bar{\beta} = \frac{\sum_i^N \frac{\beta_i}{\sigma_i^2}}{\sum_i^N \frac{1}{\sigma_i^2}} \quad (4.1)$$

With the uncertainty

$$\sigma_{\bar{\beta}} = \left(\sum_i^N \frac{1}{\sigma_i^2} \right)^{-1/2} \quad (4.2)$$

In Figure 4.2 the data points deviate close to 90 % from the CLASS power spectrum, which is similar to the difference observed, when comparing the power spectrum for various seeds. The average values are initially above the power spectrum of CLASS. Then throughout the simulation the data set for the larger simulation approaches the CLASS plot line, while the power spectrum for the smaller simulation seems to oscillate about the plot line. Additionally the data points begin to bend at smaller wavenumbers. As mentioned earlier, a simulation with a finite number of particles is prone to irregularities in the matter density, which could explain the deviation.

4.3 Calculating the Uncertainty

The power spectrum is inferred by counting the number of particles on a finite shell in the reciprocal lattice and therefore is therefore prone to error. During analysis the focus will lay on the relative difference between perturbed and unperturbed simulations. In order to compare the relative power spectra for different perturbations, it can be advantageous to first fit a curve to the data points to reduce the noise. Because all simulations have the same initial displacements of particles, the statistical scattering in the power spectra is correlated. Therefore, simply propagating the individual uncertainties to get the uncertainty for the relative power spectrum would give a value far too large. On the other hand, assuming the data points are fully correlated is not correct either.

Instead, one can apply a signal filter to the quotient to block high frequency noise and calculate the uncertainty from the difference between filtered and unfiltered data. A Gaussian filter works by convoluting the data points as done in 4.3. Here g is the Gauss function defined in 4.4.

$$Pk_{filter}(k, \sigma) = \sum_{n=-k}^{N-k} Pk(k-n)g(n, t) \quad (4.3)$$

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \quad (4.4)$$

In 4.4 σ determines how much adjacent points influence the filtered value (for $\sigma \ll 1$ the value would remain the same and for $\sigma \gg N$ every data point would equal the mean of the data set). The constant on the right hand side is a normalizing factor. In 4.3 $f(k)$ is the Pk -value for the k^{th} data point and N is the total amount of data points. Since the Gaussian function is greater than zero for all real numbers, it is common practice to truncate the function at 5σ left and right of the peak. The normalizing factor would need to be adjusted accordingly. One drawback of the Gauss filter is that the data points towards the edge of the set can not be smoothed adequately, since they are missing the adjacent point to one side. Nevertheless, this defect becomes negligible for large sets of data.

The effect that the smoothing algorithm has can be seen in Figure 4.3 for $\sigma = 4$. After applying the filter, the standard deviation could be calculated from the

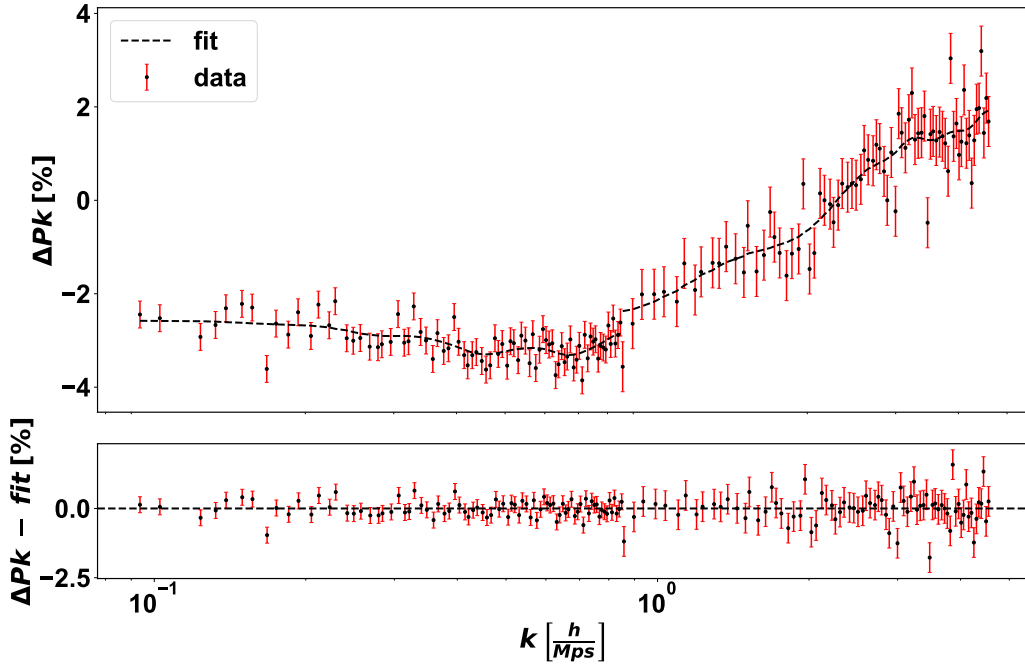


Figure 4.3: Smoothing of data set with a Gaussian filter ($\sigma = 4$). There is a larger gap at the border between the two simulations for the smoothed data, because the Gauss filter does not work well for data points towards the edge.

distance between data points before and after smoothing (bottom of 4.3). Since the noise is quite clearly affected by the simulation size, the statistical uncertainty is obtained separate for each simulation (red error-bars in 4.3). The

relative uncertainty of the relative power spectrum ranges from 10^{-3} to 0.05.

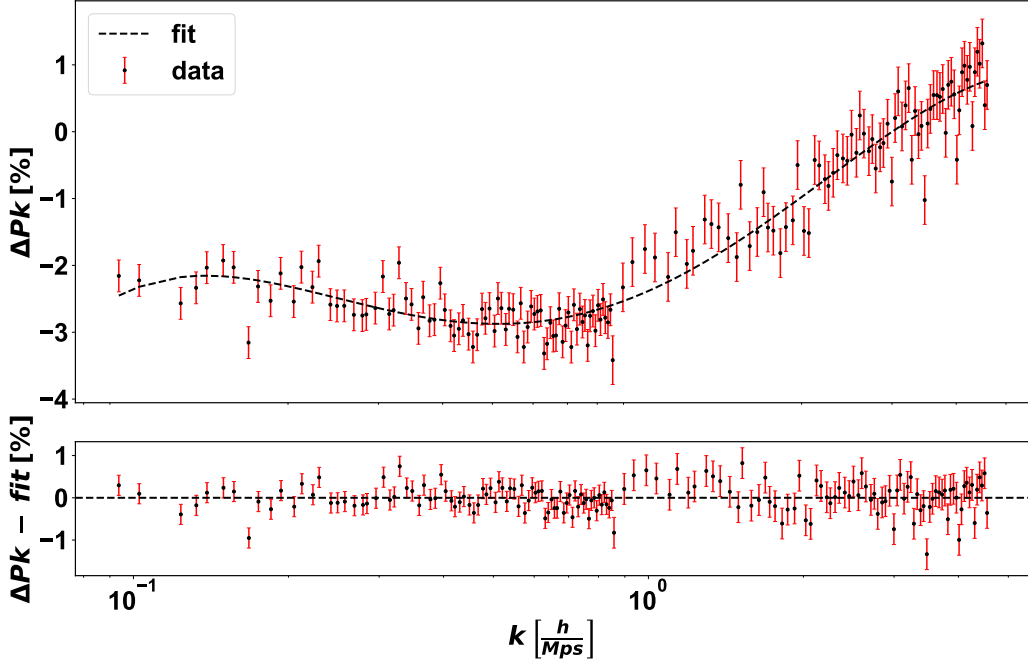


Figure 4.4: Fitting a polynomial of degree 11 to the data

It should be noted that the wavenumbers also have a statistical uncertainty (calculated by the *gevolution* code), that is at most one percent of the wavelength and is identical for perturbed and unperturbed power spectra.

After assigning an uncertainty to the data set, the next step will be to fit a hypothetical curve that can be used for better analysis. Since the power spectra seem to follow an arbitrary path (especially at small scales), a polynomial fit of varying degree will offer enough flexibility. The fit is done for the logarithm of wavenumbers. The order of the polynomial was chosen to be between 3 and 8, depending on the amplitude of the perturbation and the redshift. This allows for enough flexibility without overfitting the data. The goodness of fit would range from 1.2 to 1.6. An example can be seen in Figure 4.4

5 Modifying the Gravitational Constant

5.1 Change in Power Spectrum

After subtracting the unperturbed power spectrum, estimating the statistical uncertainty and fitting a theoretical curve we arrive at the relative power spectrum seen in Figure 5.3 for $z = 0$ and 5.1 for $z = 3$.

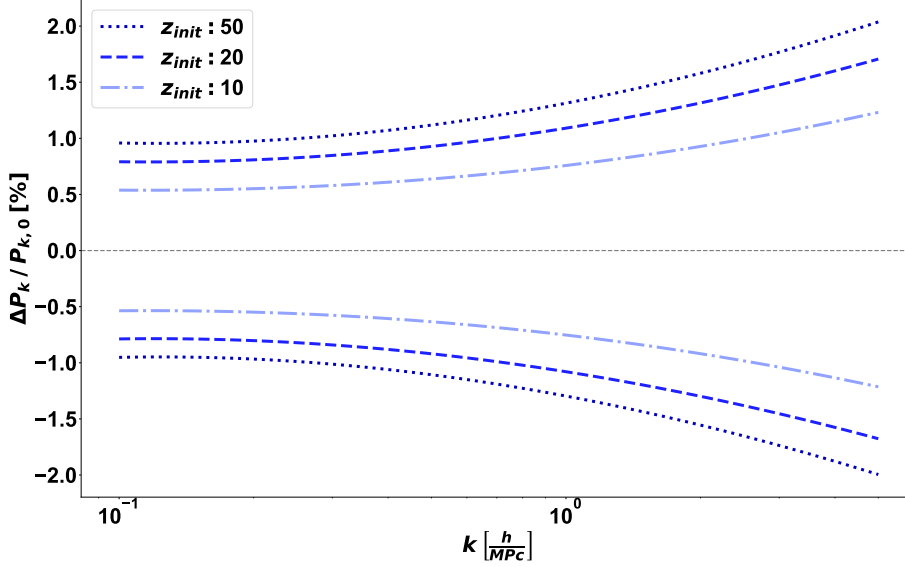


Figure 5.1: Relative power spectrum for $z = 3$. z_{init} refers to the redshift at which the modification is introduced. Although there is only one label, two curves correspond to the same value for z_{init} , where the only difference is the sign of δG

In the figures, the relative difference of the perturbed and unperturbed power spectra are shown for both positive and negative modifications. For $a(t_0) = 1$ the relative change of the gravitational constant $\delta G / G$ is merely 3×10^{-5} at $z = 50$ and 1.6×10^{-3} at $z = 0$. The power spectra for $\pm \delta G$ are almost a mirrored image at the x -axis. Differences that do occur are well within the statistical error. Because of this, I will only refer to the positive δG for analysis, as the opposite would apply to the negative values.

Firstly, one can see that in 5.1 the curves increase for smaller scales. The increase in gravitational attraction causes larger structure to form earlier. This trend is less pronounced for smaller z_{init} , since the modification happened at a later time.

Interestingly, the most significant change out of all power spectra occurs when the modification is introduced at $z = 0.3$ in 5.3 (dark red PS). Here, barely any

change is detectable for smaller wavenumbers, meaning the formation of larger structures remains the same. On the other hand, the stronger gravitational interactions affects the distance of particles at smaller scales. This may be related to the following example.

When describing a mass orbiting around a much larger mass, the radial range of the smaller mass is determined by the gravitational force and kinetic energy. In Figure 5.2, the effective potential is displayed for different gravitational constants. Particles with the same maximal kinetic energy are on average closer in radius, when exposed to a larger gravitational force (outlined in purple).

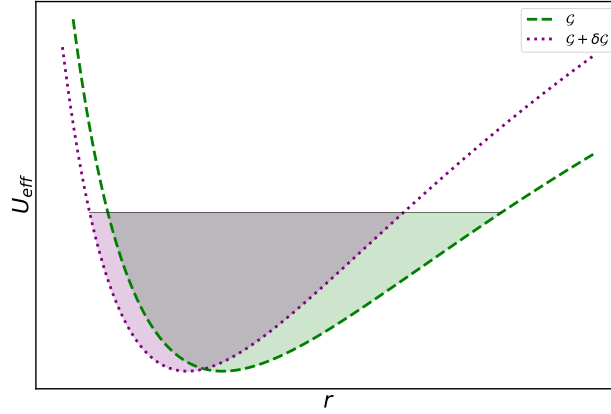


Figure 5.2: U_{eff} for different gravitational constants, where $\delta\mathcal{G} > 0$. When the gravitational constant is larger, the smaller mass is closer to the larger mass.

For the larger z_{init} in Figure 5.3 a peak shows up at approximately $0.5 [\frac{h}{Mpc}]$ and has no significant change at smaller ranges. Since the modification was introduced at earlier times, the universe was nearly homogeneous. Thus, an increased gravitational force results in clusters to attract more particles, that would otherwise escape the gravitational attraction and therefore larger structures form.

The power spectrum lays above the unchanged power spectrum for all wavenumbers indicating the overall increase of the overdensity for $k > 0.1 [\frac{h}{Mpc}]$.

In Figure 5.4, the power spectra are displayed for different redshifts ranging from 1.5 to 0. The colors correspond to the redshift z_{init} at which the change was done. When the change was introduced before $z = 1$ the power spectra are almost identical in shape although with different amplitudes. Interestingly, the data sets increasingly oscillate with time. Paying attention to the purple line between $z = 1.5$ and $z = 1$, the following analogy could lend intuition for the increased oscillations.

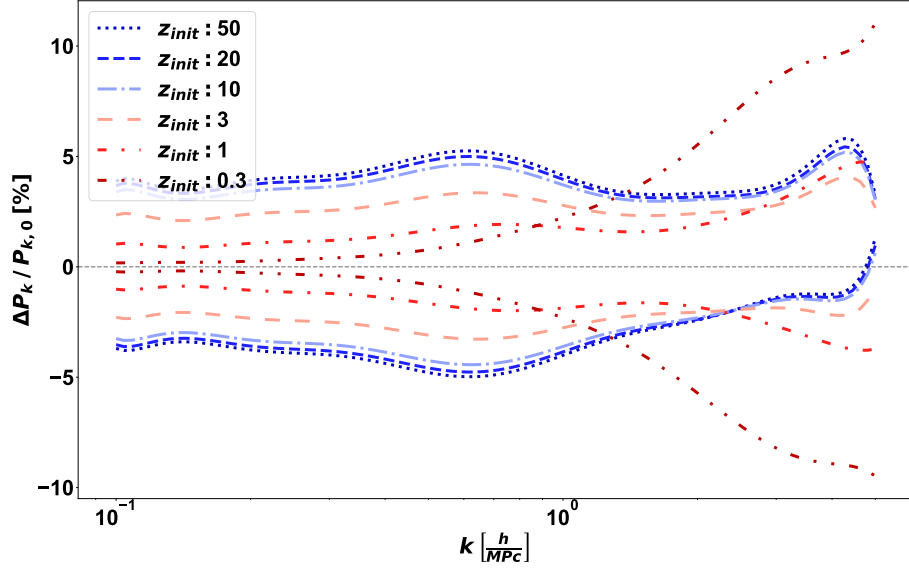


Figure 5.3: Relative power spectrum for $z = 0$. The power spectra that began at earlier times are not symmetric at the x -axis for opposite signs.

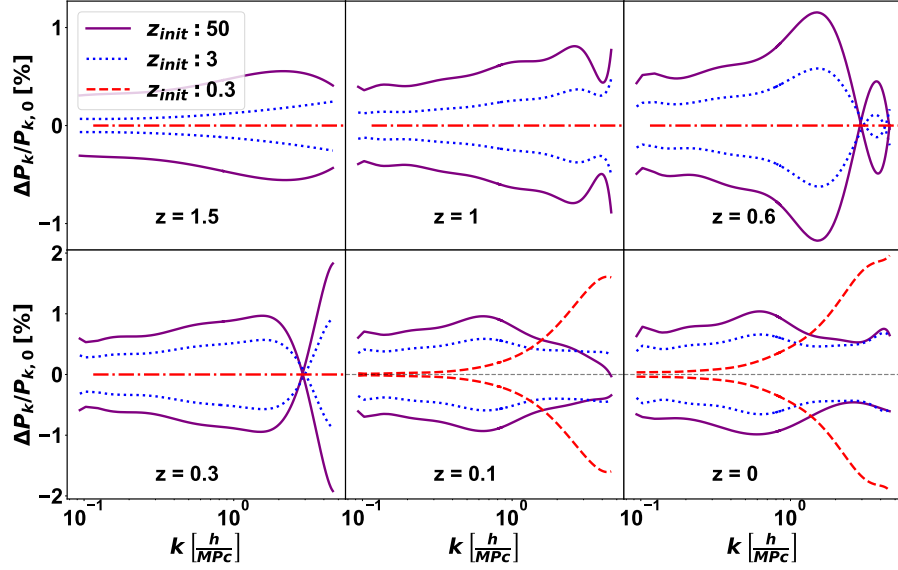


Figure 5.4: Relative power spectra for various redshifts after changing the gravitational constant.

In the initial configuration, massive bodies with equal mass are aligned along a one dimensional chain with equal distance between each body. Now, if every mass is moved by a small distance along the chain where the direction is

switched after every mass, then the masses would begin to move to the neighboring body through gravitational attraction. As mentioned before, the power spectrum describes the typical amplitude of a mode in the correlation function. At equal distance between the bodies, the power spectrum would show one peak at the corresponding wavenumber. While the masses move towards each other the peak would split up in to two separate peaks since we now have two characteristic distances. Since gravity affects how much the bodies accelerate, the formation of the two peaks also relates to the gravitational constant. Back in Figure 5.4 the oscillations in the power spectrum could therefore be the result of one mode splitting up into two or more modes at an accelerated rate due to the perturbation.

5.2 Comparing Different Perturbations

Next, the linear relationship between power spectra with different perturbation will be examined. From the output for two different perturbations one can calculate the ratio of relative change according to

$$\frac{Pk_{\delta G,2} - Pk_0}{Pk_{\delta G,1} - Pk_0} \quad (5.1)$$

where $Pk_{\delta G,i}$ is the power spectrum for a perturbed simulations with the factor δG_i and Pk_0 is the unperturbed data. The perturbation of the power spectrum in the denominator of equation 5.1 was chosen to be $\delta G_1 = \pm 0.5$ (in actuality it is equal to $\pm 0.5 \times 10^{-16}$, but since the absolute scale is not important, it will from now on be referred to as ± 0.5), which corresponds to a relative change of $\delta G/G = \pm 7.89 \times 10^{-5}$. The numerator was varied for

$$\delta G_2 \in \{\pm 1, \pm 5, \pm 10, \pm 20\}$$

where the sign matches the sign of the denominator. The overall trend can be seen quite clearly in Figure 5.5, because high frequency noise was filtered.

The resulting values are close to the ratio of the perturbations. At $k > 1 [\frac{h}{Mpc}]$ the data points begin to deviate from the straight line, meaning that the perturbations are likely too large for small scales and thus not linear. Interestingly, the values are systematically above the ratio for positive and below for negative perturbations (dotted line in Figure 5.5), which increases for larger $|\delta G|$. The mean values and their uncertainties are calculated from the data points with a wavenumber less than $0.7 [\frac{h}{Mpc}]$ and shown in Table 5.1 The percent change that a negative perturbation invokes in the power spectrum decreases for smaller scales. The reason for this can be understood when looking at Figure 5.6. The power spectra with larger constants oscillate throughout time at smaller scales

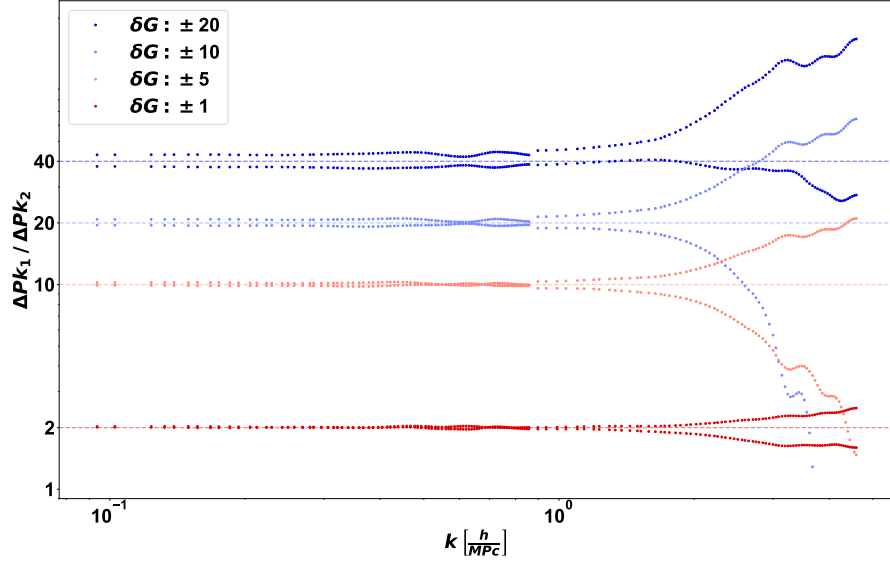


Figure 5.5: Ratio of relative change for $Pk_{\delta G,1} = 0.5$ and various $Pk_{\delta G,2}$ as described in equation 5.1. The two sets with matching colors belong to a perturbation with $\pm\delta G_2$, where the values that increase for larger k correspond to $+\delta G_2$ and decrease for negative changes. The data has been smoothed by the Gauss filter stated in section 4.3

δG_2	Mean Value	rel. Diff. [%]
-20	37.47 ± 0.14	6.53 ± 0.35
20	43.21 ± 0.19	7.72 ± 0.48
-10	19.456 ± 0.062	2.76 ± 0.31
10	20.682 ± 0.079	3.35 ± 0.39
-5	9.927 ± 0.031	0.73 ± 0.0031
5	10.126 ± 0.037	1.25 ± 0.0037
-1	2.0149 ± 0.0069	0.74 ± 0.35
1	1.9961 ± 0.0073	0.20 ± 0.36

Table 5.1: Mean values of $\Delta Pk_1/\Delta Pk_2$ calculated for wavenumbers less than $0.7[\frac{h}{Mpc}]$, because the data points begin to deviate from the straight line beyond this point. In the third column the relative difference to $\delta G_2/\delta G_1$ is displayed in percent.

and redshifts.

In conclusion, the change resulting from a gravitational perturbation exhibits a linear relationship. So far, the modification had always been implemented at the beginning of the simulation ($z = 50$).

This trend is consistent even when the modification was not introduced at

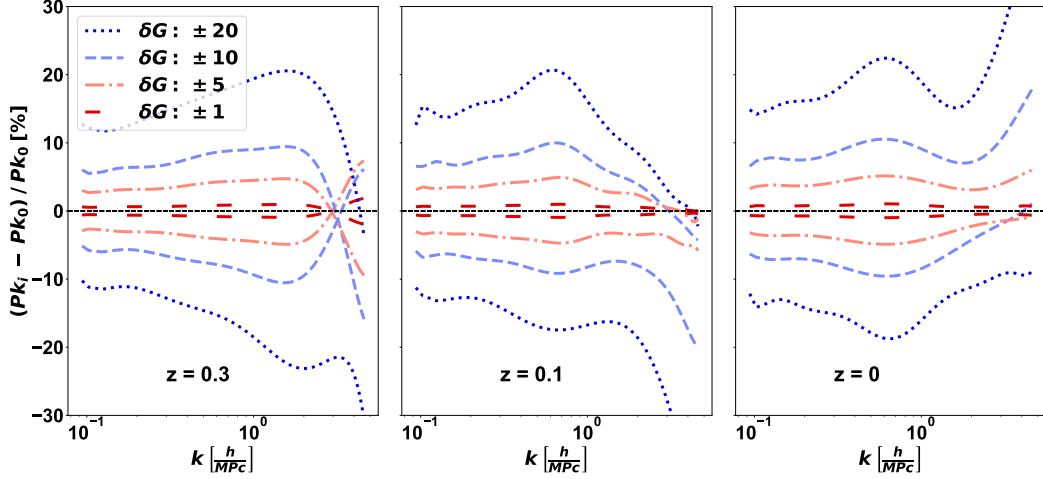


Figure 5.6: Relative change of power spectra for different modifications. The data sets with larger δG oscillate for smaller wavenumbers. The modification was introduced at the beginning of the simulation.

$z = 50$. When, for example, one simulation has $\delta G_{10} = 10$ and another has $\delta G_5 = 5$, where both modifications were introduced at $z = 10$, then the change in the power spectrum at $z = 0$ is double (for a sufficiently small k). If the larger perturbation is instead introduced at $z = 1$, the relative power spectra would still remain similar in shape, but the difference in amplitude would now be smaller.

In the following section, we will try to obtain a more rigorous understanding of the relationship between the ratios of the relative power spectra and the time at which the modification was submitted.

There are three values uniquely identifying a power spectra, which are the perturbation (δG), the redshift at which the perturbation was initialized in the simulation (z_{init}) and the redshift at which the power spectra was calculated. For now, the latter will be set to $z = 0$. This means that there are 4 degrees of freedom for two power spectra. If both perturbations are set constant, as well as one of the z_{init} , then this leaves one variable. To avoid confusion, $z_{\text{init},2}$ for simulation 2 is to be kept the same, while in simulation 1 $z_{\text{init},1}$ will be varied

The mean values and their uncertainties for the ratio of the relative power spectra are calculated the same way as the values in Table 5.1. This is repeated for different $z_{\text{init},1}$ and the resulting values are fitted against $z_{\text{init},1}$ according to

$$\frac{\delta G_1}{\delta G_2} (\beta_0 + \beta_1 a + \beta_2 a^2 + \beta_3 a^3) \quad (5.2)$$

Where $a = (1 + z_{\text{init},1})^{-1}$ is the scale factor. The library `scipy.odr` [2] is used for the fit. Fitting the data points was done for all unique combinations of two simulations with the parameter

- $\delta G \in \{20, 10, 5, 1, 0.5\}$
- $z_{\text{init},2} \in \{50, 20, 10, 3, 1, 0.3\}$

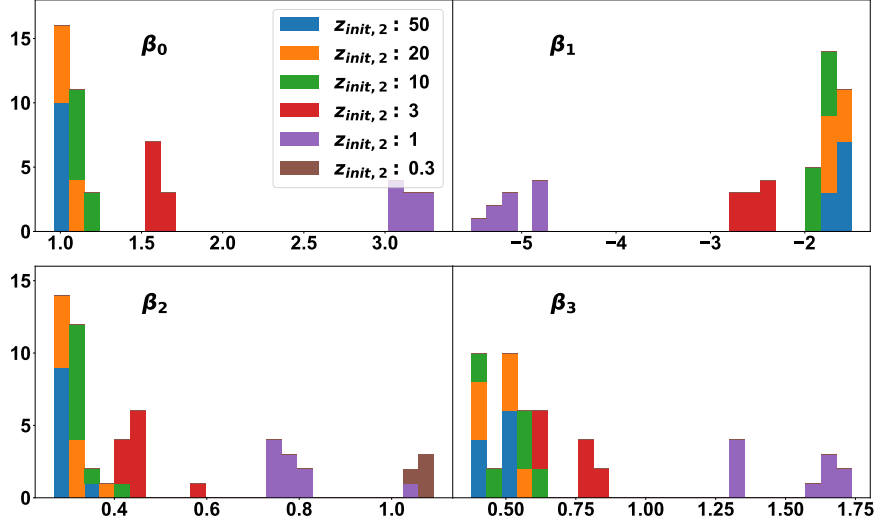


Figure 5.7: Parameters of the fitted curve in 5.2 for different $z_{\text{init},2}$. The values for smaller redshifts are increasingly further apart. The histogram for negative perturbations looks very similar

In Figure 5.7, the different values for β_i can be seen. The different colors correspond to $z_{\text{init},2}$. One may notice that the entries for $z_{\text{init},2} = 0.3$ are missing, which is because they are beyond the borders of the x -axis. The increased distance between the values for smaller redshift is a distinguishing trait for all parameter. It is the result of a larger factor than $\delta G_1 / \delta G_2$ in equatio 5.2. This has to do with the fact that if $\delta G_1 > \delta G_2$ and the perturbation for simulation 2 was introduced at a smaller $z_{\text{init},2}$, then the difference between the two relative power spectra would grow. Instead of using the function 5.2 it can be rearranged according to

$$\frac{\delta G_1}{\delta G_2} \beta_0 (1 + \beta_1 a + \beta_2 a^2 + \beta_3 a^3) \quad (5.3)$$

The result is that the parameters are significantly closer in value, except for β_0 , which still varies based on $z_{\text{init},2}$.

Again looking at Table 5.1 the largest systematic deviation is for $\delta G = \pm 20$. To obtain a more accurate mean value for the parameter only perturbations smaller than 20 were included. Since all values for the parameter have an error

the mean value was calculated using the weighted average (equations 4.1 and 4.2). One should note that the weighted average is only valid if the data points are within one σ of each other. In this case they are. The parameter can be seen in Table 5.2 for both positive and negative perturbations (The fit is only done if both δG_1 and δG_2 are either positive or negative).

	$+\delta G$	$-\delta G$	average
β_1	-1.6363 ± 0.0033	-1.7436 ± 0.0017	-1.6899 ± 0.0018
β_2	0.25701 ± 0.011	0.39201 ± 0.0054	0.32451 ± 0.006
β_3	0.51714 ± 0.0085	0.49421 ± 0.0043	0.50568 ± 0.0048

Table 5.2: Mean value of parameters for both positive and negative perturbations and the average of the two values

If the systematic shift discussed earlier is equal in amplitude for both positive and negative perturbations then the average of the two would eliminate the systematic error. The systematic shift is slightly larger for negative δG , meaning the average of the two would not fully erase the systematic error.

6 Modifying the Hubble Parameter

6.1 Change in Power Spectrum

If $\delta\mathcal{H} = 10^{-5}$, it results in a relative perturbation of $a \cdot \delta\mathcal{H} / \mathcal{H} = 10^{-6}$ at $z = 50$ and 0.03 at $z = 0$. For more clarity, we will from now on refer to $\delta\mathcal{H}$ in units that are 6 orders in magnitude smaller. For example, a perturbation of 10^{-5} would be 10.

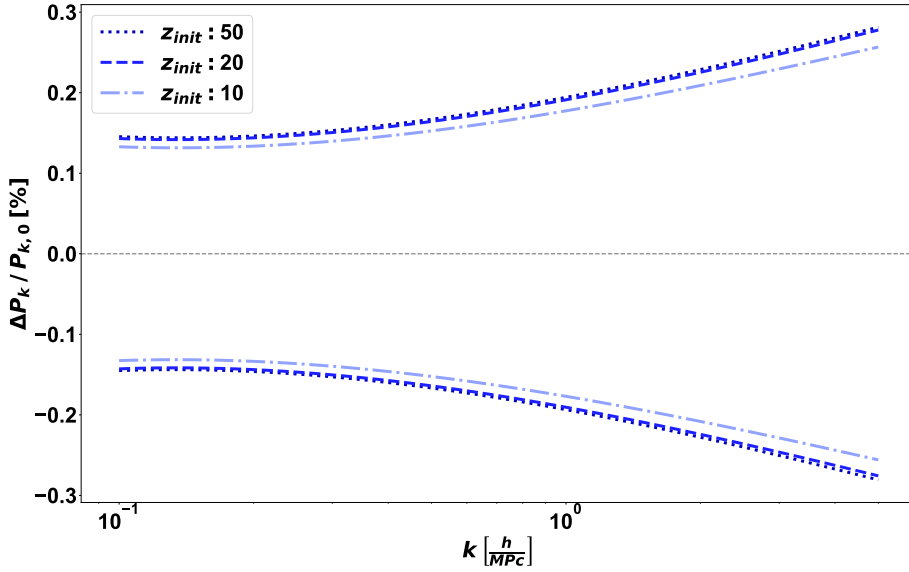


Figure 6.1: Relative power spectra for different starting points (z_{init}) at $z = 0$. The outcome is similar as when the gravitational constant is changed

The relative power spectra can be seen in figure 6.1 for $z = 3$ and 6.2 for $z = 1$. The Hubble parameter was changed at different times during the simulation. There are two sets of data above and below the x -axis that correspond to one label with the only difference being the sign of the modification $\pm\delta\mathcal{H}$. As shown in Figure 5.1, each pair of power spectra with opposite sign $\pm\delta\mathcal{H}$ are almost mirrored images at the x -axis, although this time the power spectrum for positive $\delta\mathcal{H}$ is below the x -axis. The sets of data for different z_{init} are much closer to each other than the data sets for the modified gravitational constant. An explanation for this may be the following; when adding $\delta G a$ to the gravitational force, the change in force scales asymptotically with r^{-2} as particles approach each other, while the change in acceleration due to the perturbation of the Hubble parameter is nearly constant.

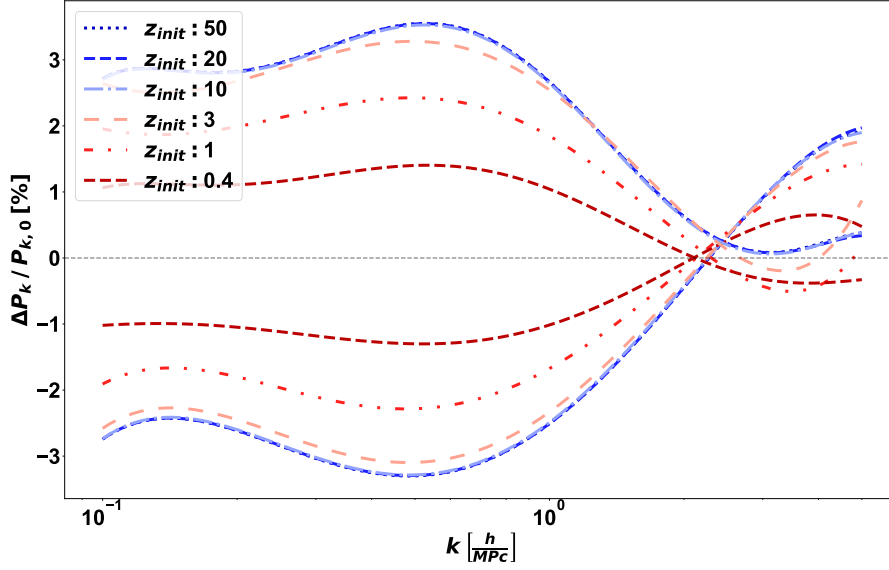


Figure 6.2: Relative power spectra at $z = 0$. The power spectrum no longer behaves as if G were changed

In Figure 6.2, the change in power spectra is the largest at about $0.55 [\frac{h}{Mpc}]$. Perhaps the gravitational force will counteract the additional expansion at smaller scales, where the gravitational force is stronger. This time for all power spectra, there is a clear peak at around $0.5 [h/Mpc]$.

The simulations were repeated for different perturbations. When comparing the relative power spectra to each other, they feature the same shape with varying amplitudes if $|\delta\mathcal{H}| < 100$. For a modification equal to 100, non-linear contributions are noticeable (Figure 6.3). Similar to δG , the power spectra begin to oscillate in time at smaller scales. Since this trend has occurred for both the gravitational and Hubble modification, the fluctuations could be the result of the systematic error from the initial particle displacements discussed in Chapter 4.2. In Figure 4.2, the power spectrum oscillates about the CLASS data points for larger wavenumbers. This trend may be enhanced by the perturbations. One could test this hypothesis with a larger simulation.

6.2 Relationship between Different Perturbations

After dividing the relative power spectra (according to equation 6.1), the data points scatter around a horizontal line located at $\delta\mathcal{H}_1 / \delta\mathcal{H}_2$.

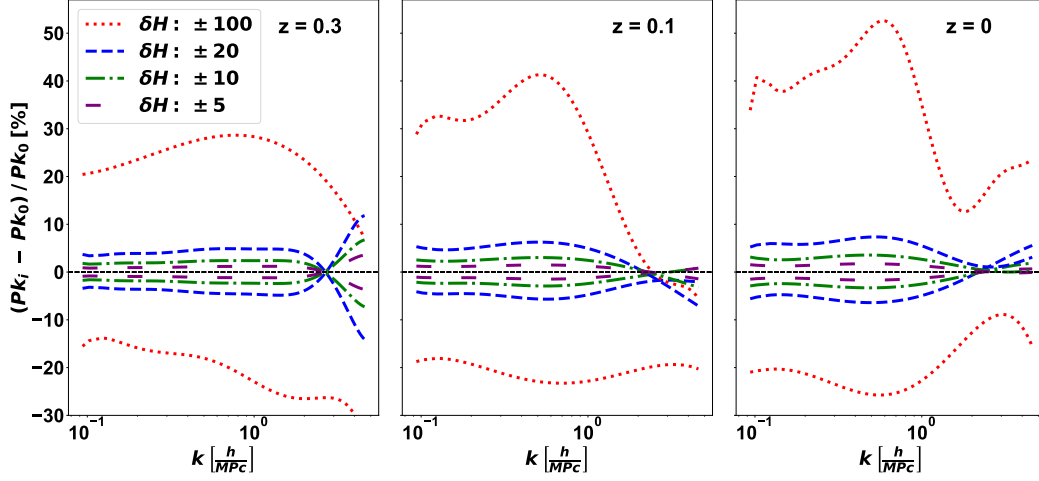


Figure 6.3: Difference in power spectra for different perturbations. The data set for $\pm\delta\mathcal{H} = 100$ is no longer linear. Even at smaller perturbations, the power spectra are systematically below the unperturbed power spectrum at $z = 0.1$ and above at $z = 0$

$$\frac{Pk_{\delta\mathcal{H},1} - Pk_0}{Pk_{\delta\mathcal{H},2} - Pk_0} \quad (6.1)$$

At $k = 0.7$, the data points deviate from the horizontal line. The mean value and standard deviation for the ratios is calculated for wavenumbers smaller than $0.7 [\frac{h}{Mpc}]$. The calculated values can be seen in Table 6.1. Again, there is

$\delta\mathcal{H}_1$	Mean Value	rel. Diff. [%]
100	142.93 ± 0.99	42.93 ± 0.99
-100	78.81 ± 0.37	21.19 ± 0.37
20	20.890 ± 0.056	4.45 ± 0.28
-20	19.323 ± 0.063	3.38 ± 0.32
10	10.136 ± 0.027	1.35 ± 0.27
-10	9.96 ± 0.03	0.41 ± 0.30
5	4.977 ± 0.014	0.47 ± 0.28
-5	5.074 ± 0.014	1.48 ± 0.29

Table 6.1: Mean values of $\Delta Pk_1 / \Delta Pk_2$ calculated for wavelengths smaller than $0.7 [\frac{h}{Mpc}]$ because the perturbation is not linear beyond this point. The third column shows relative difference of $\delta\mathcal{H}_1/\delta\mathcal{H}_2$, displayed in percent

a systematic shift that increases for larger perturbations and is the result of the fluctuations of the power spectrum discussed earlier.

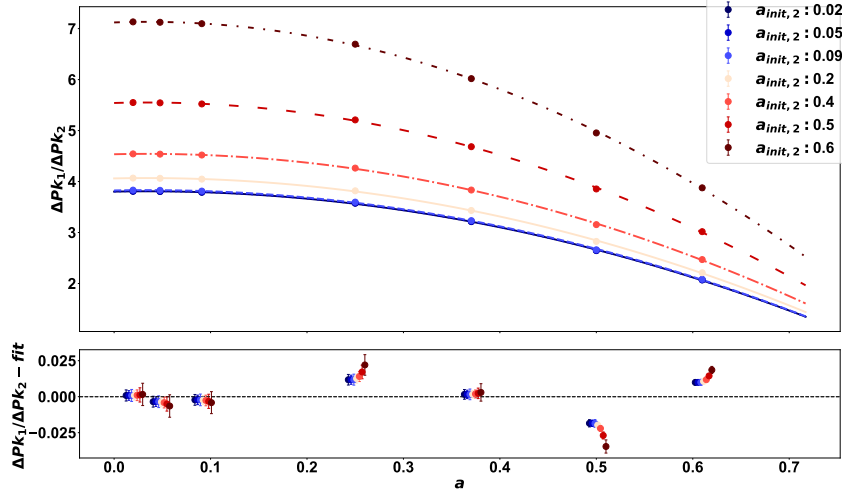


Figure 6.4: Theoretical curve in equation 6.2 fitted to the data points for different scale factors. The perturbation of the simulation is 20, while it is 5 for the second. Therefore, if both simulations begin at the same scale factor the quotient will remain at 4. In the lower plot, the theoretical value is subtracted from the data. It shows that the data deviates less than 0.01 from the polynomial.

By varying z_{init} , for one of the power spectra the resulting values can be fitted against z_{init} with the function in equation 6.2. This is done to verify if the linear relationship still remains when varying z_{init} .

$$\frac{\delta \mathcal{H}_1}{\delta \mathcal{H}_2} \beta_0 (1 + \beta_1 a + \beta_2 a^2 + \beta_3 a^3) \quad (6.2)$$

which apart from the constant is the same function used when the gravitational constant was modified. Here, a refers to the scale factor at which the perturbation of simulation 1 was introduced. Notice the parameter β_0 is outside of the parenthesis, the reason for this will become more clear later on.

The parameters in simulation 2 will remain constant. In simulation 1, the redshift at which the modification is initiated ($z_{init,1}$) will be varied. Again, the data points were fitted using the **scipy odr** [2] library.

The theoretical curve can be seen in Figure 6.4 where the x-axis denotes the scale factor at which the modification was introduced in simulation 1. The different data sets correspond to different scale factor for which the change was introduced in simulation 2. The perturbation $\delta \mathcal{H}$ for simulation 1 is 20, while for the second simulation $\delta \mathcal{H} = 5$. Due to the linear relationship whenever the perturbation was initiated at the same scale factor, the quotient will be close to 4. The goodness of fit is close to one. The displacements of the data points to the polynomial are correlated for different $a_{init,2}$ (6.4 lower plot). This may be due to the following; As seen in Figure 6.2, the power spectra deviate

systematically from the unperturbed data set. The important thing to note is that this effect is enhanced if the perturbation is invoked earlier during the simulation. Therefore the longer simulation 2 remains unperturbed the larger the gap between the two simulations would become. This is the case in Figure 6.2.

Next, all unique combinations for two power spectra are fitted to determine the parameter β_i for all cases. The possible values for $\delta\mathcal{H}$ and z_{init} are

- $\delta\mathcal{H}$: $\pm 20, \pm 10, \pm 5, \pm 1$
- z_{init} : 50, 20, 10, 3, 1.7, 1, 0.64

Where the sign of δH_i will remain the same because of the systematic error mentioned earlier. A perturbation of ± 100 was not included as it is not linear. Interestingly, all parameters β_i are within one standard deviation apart from each other, except for β_0 , where the values vary depending on the parameter $\delta\mathcal{H}$ and z_{init} . This gives some evidence to the fact that $\delta\mathcal{H}$ and z_{init} can be summarized as one variable.

From the parameter β_i the mean value and statistical uncertainty can be calculated as seen in equations 4.1 and 4.2. The calculated values are listed in Table 6.2. Compared to the values calculated when changing G , the difference for

	$\delta\mathcal{H}$	$\delta\mathcal{H}$	average
β_1	0.17869 ± 0.0099	0.17877 ± 0.0098	0.1787 ± 0.0070
β_2	-1.755 ± 0.039	-1.7542 ± 0.039	-1.755 ± 0.028
β_3	0.384 ± 0.040	0.3935 ± 0.039	0.389 ± 0.028

Table 6.2: The parameter for the theoretical curve fit for both positive and negative perturbations of the Hubble parameter. The average is shown in the fourth column

positive and negative perturbations is much smaller, as well as their respective uncertainties. Also the polynomial is concave while for G it is convex, going back to the observation, that the rate of change is more rapid for gravity

Using the calculated parameter, we can calculate at what scale factor a perturbation would need to be introduced to achieve the same outcome as if a perturbation with half the value was introduced at $z = 50$.

$$\frac{2(1 + \beta_1 a + \beta_2 a^2 + \beta_3 a^3)}{1 + \beta_1(1 + 50)^{-1} + \beta_2(1 + 50)^{-2} + \beta_3(1 + 50)^{-3}} = 1$$

$$\implies a \approx 0.64$$

or for redshift

$$z \approx 0.57$$

7 Conclusion

The similarities in outcome when modifying the gravitational constant or the Hubble parameter are numerous. For one, the relative power spectra for early modifications show a maximum at around $0.5 [\frac{h}{Mpc}]$ and then decrease for larger wavenumbers. Furthermore, the relative power spectrum shows a systematic displacement at smaller scales and larger perturbations, which is likely caused by the random number generator used for the initial configuration of the particles. On the other hand there are plenty of discrepancies between the two parameter modifications. For example, when comparing the change induced at different z_{init} the relative power spectra for $\delta\mathcal{H}$ are much closer in amplitude than when gravity is modified. This is because the gravitational force is proportional to $\mathbf{r}^{-2}$ and thus changes more rapidly than the Hubble parameter, that only depends on the scale factor and its derivative, which are global variables.

For each parameter, we were able to infer a linear relationship for perturbation and relative difference between perturbed and unperturbed power spectrum. The perturbation at which non linear factors became significant is at $\delta G = 20$ or a relative change of $\delta G / G = 3.16 \times 10^{-3}$. For the Hubble Parameter, the case is the same for $\delta\mathcal{H} = 100$ or a relative difference of $\delta\mathcal{H} / \mathcal{H}(a=1) = 0.3$ at the present time.

Lastly, we looked at the effect on the change of the power spectrum when altering the perturbation or time at which the perturbation was incurred. By changing z_{init} for one simulation and comparing it to a simulation that was kept constant, we could establish a relationship between percent change of the power spectrum and z_{init} . The result was that the same change in power spectrum at $z = 0$ can be achieved, with various combinations for δG and z_{init} (or for $\delta\mathcal{H}$ and z_{init} , depending on the parameter that was modified). A fitting analogy to this would be a linear function

$$f(x) = a(x - b),$$

with the requirement

$$f(x_0) = y_0$$

where there are infinitely many combinations of a and b that would fulfill the requirement.

As a continuation to this paper, it would be interesting to conduct the same calculations for simulations with other seeds to quantify the systematic error that is seen for smaller wavenumbers. One can calculate the relative difference between the ratio of perturbations to the mean value, as was done in Table 5.1 and compare them to different seeds.

Additionally to the linear perturbation in a it would be interesting to determine the effect it would have, when adding a constant to the parameter or a perturbation of higher order in a . Through a linear combination of multiple

perturbations it one can determined more precisely the rate of change induced at different times during the simulation in the power spectrum.

Appendix

Setting	Value
lattice structure	simple cubic
baryon treatment	blend
seed	1234 ¹
correct displacement	yes
k – domain	sphere
k _{pivot}	0.05
A _s	2.215×10^{-9}
n _s	0.9619
h	0.67556
omega _b	0.022032
omega _{cdm}	0.022032
T _{cmb}	2.7255
N _{ur}	3.046
initial redshift	50
boxsize	[150, 28]
Ngrid	256
Courant factor	48.0
time step limit	0.04
gravity theory	Newton
vector method	paraboloc
Pk redshifts	[50, 10, 3, 1, 0.4, 0]
Pk outputs	rhoN
Pk bins	1024

Table A.1: *gevolution* settings. Some settings regarding output files are omitted. For more information on the settings meaning refer to [4]

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